

```
path = "/content/ecommerce_sales_data (2).csv"
import pandas as pd
df= pd.read_csv(path)
df
```

	Order Date	Product Name	Category	Region	Quantity	Sales	Profit
0	2024-12-31	Printer	Office	North	4	3640	348.93
1	2022-11-27	Mouse	Accessories	East	7	1197	106.53
2	2022-05-11	Tablet	Electronics	South	5	5865	502.73
3	2024-03-16	Mouse	Accessories	South	2	786	202.87
4	2022-09-10	Mouse	Accessories	West	1	509	103.28
...	...	...	...	...	...	...	...
3495	2023-02-15	Monitor	Accessories	North	4	4064	771.16
3496	2022-09-18	Monitor	Accessories	East	1	1117	119.89
3497	2022-04-12	Laptop	Electronics	South	4	260	66.02
3498	2022-01-18	Printer	Office	South	3	222	50.28
3499	2024-11-01	Smartwatch	Electronics	East	9	10530	1777.32

3500 rows × 7 columns

Double-click (or enter) to edit

```
import pandas as pd
df= pd.read_csv("/content/ecommerce_sales_data(2).csv")
df
```

```
-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_2930/2357098246.py in <cell line: 0>()
      1 import pandas as pd
----> 2 df= pd.read_csv("/content/ecommerce_sales_data(2).csv")
      3 df

----- 4 frames -----
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf,
mode, encoding, compression, memory_map, is_text, errors, storage_options)
    871     if ioargs.encoding and "b" not in ioargs.mode:
    872         # Encoding
--> 873         handle = open(
    874             handle,
    875             ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory:
'/content/ecommerce_sales_data(2).csv'
```

```
df.head()
```

	Order Date	Product Name	Category	Region	Quantity	Sales	Profit
0	2024-12-31	Printer	Office	North	4	3640	348.93
1	2022-11-27	Mouse	Accessories	East	7	1197	106.53
2	2022-05-11	Tablet	Electronics	South	5	5865	502.73
3	2024-03-16	Mouse	Accessories	South	2	786	202.87
4	2022-09-10	Mouse	Accessories	West	1	509	103.28

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3500 entries, 0 to 3499
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Order Date      3500 non-null   object
1   Product Name    3500 non-null   object
2   Category        3500 non-null   object
3   Region          3500 non-null   object
4   Quantity        3500 non-null   int64
5   Sales           3500 non-null   int64
6   Profit          3500 non-null   float64
dtypes: float64(1), int64(2), object(4)
memory usage: 191.5+ KB
```

```
df.shape

(3500, 7)
```

```
df.isnull().sum()

      0
Order Date  0
Product Name  0
Category    0
Region      0
Quantity    0
Sales       0
Profit      0

dtype: int64
```

```
df.describe()
```

	Quantity	Sales	Profit
count	3500.000000	3500.000000	3500.000000
mean	4.931714	3047.966000	527.047203
std	2.575895	2440.213237	504.139732
min	1.000000	51.000000	6.970000
25%	3.000000	1049.500000	158.695000
50%	5.000000	2350.500000	361.070000
75%	7.000000	4537.000000	729.125000
max	9.000000	10782.000000	2946.930000

```
df.groupby("Category")["Sales"].sum()
```

Sales	
Category	
Accessories	4247591
Electronics	5326074
Office	1094216

dtype: int64

```
df.groupby("Region")["Sales"].sum()
```

Sales	
Region	
East	2675110
North	2488773
South	2659548
West	2844450

dtype: int64

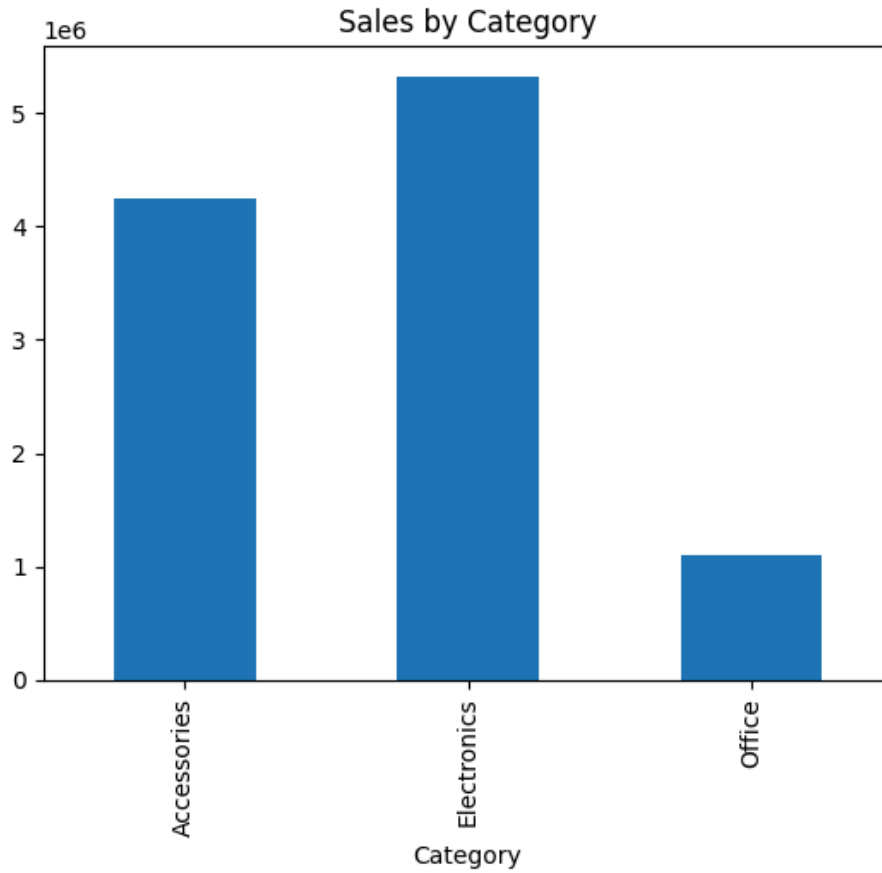
```
df.sort_values(by="Sales", ascending=False).head()
```

	Order Date	Product Name	Category	Region	Quantity	Sales	Profit
1127	2023-09-29	Keyboard	Accessories	South	9	10782	2730.14
32	2023-04-09	Laptop	Electronics	North	9	10773	1907.36
2573	2022-06-27	Laptop	Electronics	East	9	10728	1967.84
1489	2023-12-18	Monitor	Accessories	West	9	10665	1595.08
379	2022-05-09	Printer	Office	West	9	10656	961.30

```
import matplotlib.pyplot as plt
```

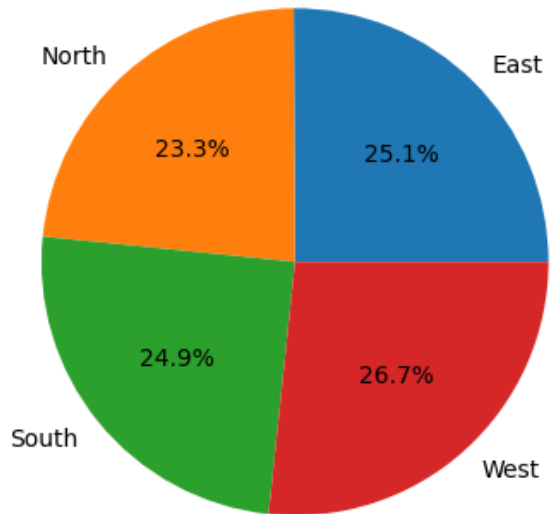
```
import matplotlib.pyplot as plt

df.groupby("Category")["Sales"].sum().plot(kind="bar")
plt.title("Sales by Category")
plt.show()
```



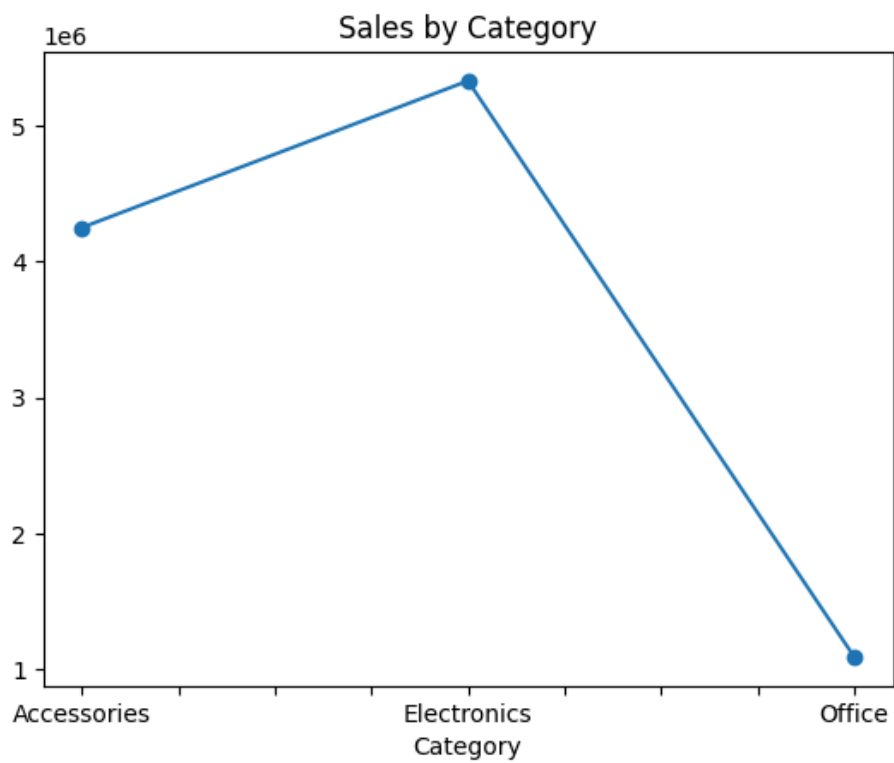
```
df.groupby("Region")["Sales"].sum().plot(kind="pie", autopct="%1.1f%%")
plt.title("Sales by Region")
plt.ylabel("")
plt.show()
```

Sales by Region



```
from google.colab import drive
drive.mount('/content/drive')
```

```
df.groupby("Category")["Sales"].sum().plot(kind="line", marker="o")
plt.title("Sales by Category")
plt.show()
```



```
df.columns
```

```
Index(['Order Date', 'Product Name', 'Category', 'Region', 'Quantity', 'Sales',  
      'Profit'],  
      dtype='object')
```

New Section